

How the vector database works

Meaning-based retrieval for the Agentic Healthcare Assistant

The vector database helps the assistant find relevant information **by meaning**, even when a question uses different wording from the stored text.

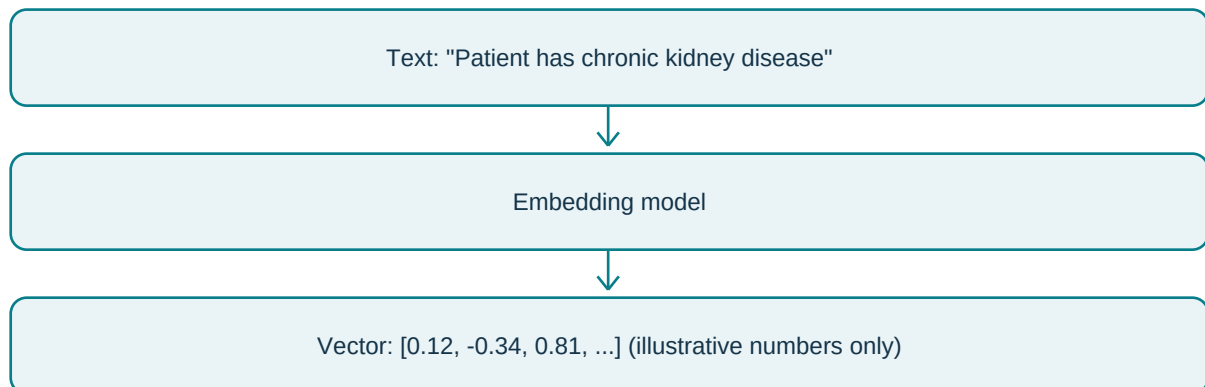
The BRD explicitly calls for a vector database such as FAISS to **store and retrieve patient summaries**, with memory modules retaining long-term patient context. It separately requires a patient database, appointment APIs and medical information search (BRD, page 3).

1. What is a vector?

An embedding model converts text into a list of numbers called a **vector** or **embedding**. Texts expressing related ideas should have relatively close embeddings in the model's mathematical space.

Stored summary: "Patient has chronic kidney disease and a history of hypertension."

Question: "What kidney problems and blood pressure history does this patient have?"



The embedding model creates the numbers; **FAISS indexes and searches them**. The original text is retained so the assistant can read the retrieved information.

Two uses in this project

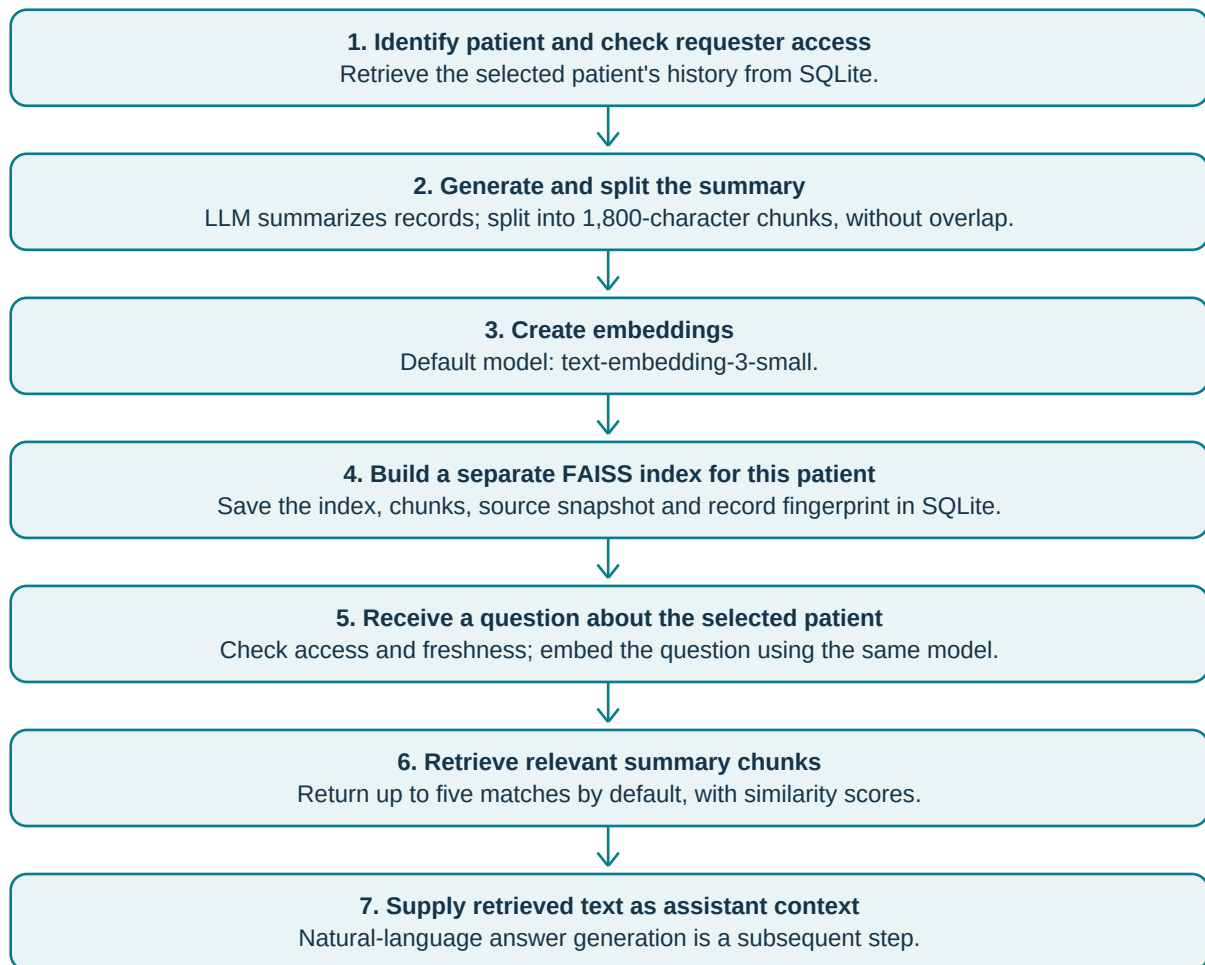
Patient-summary memory: retrieves context for a selected patient.

Shared reference RAG: retrieves relevant passages from uploaded medical PDFs.

FAISS supplies the vector-search capability. In this implementation, SQLite persists the indexes and associated text or metadata.

Patient-summary flow

The BRD requirement implemented in `src/vector_store/patient_summaries.py`



Freshness and access checks

The **rebuild()** method retrieves history, generates a summary and indexes it. Reindexing replaces the patient's whole index, preventing duplicate accumulation.

If medical records change, search returns **stale** with no matches; the summary needs rebuilding. If no summary index exists, it returns **missing**. Access is checked through the patient-history retrieval path.

Vector similarity does not determine which patient the requester may access. Search operates on the selected patient's index.

Similarity and reference RAG

How retrieval works, and how retrieved evidence reaches the LLM

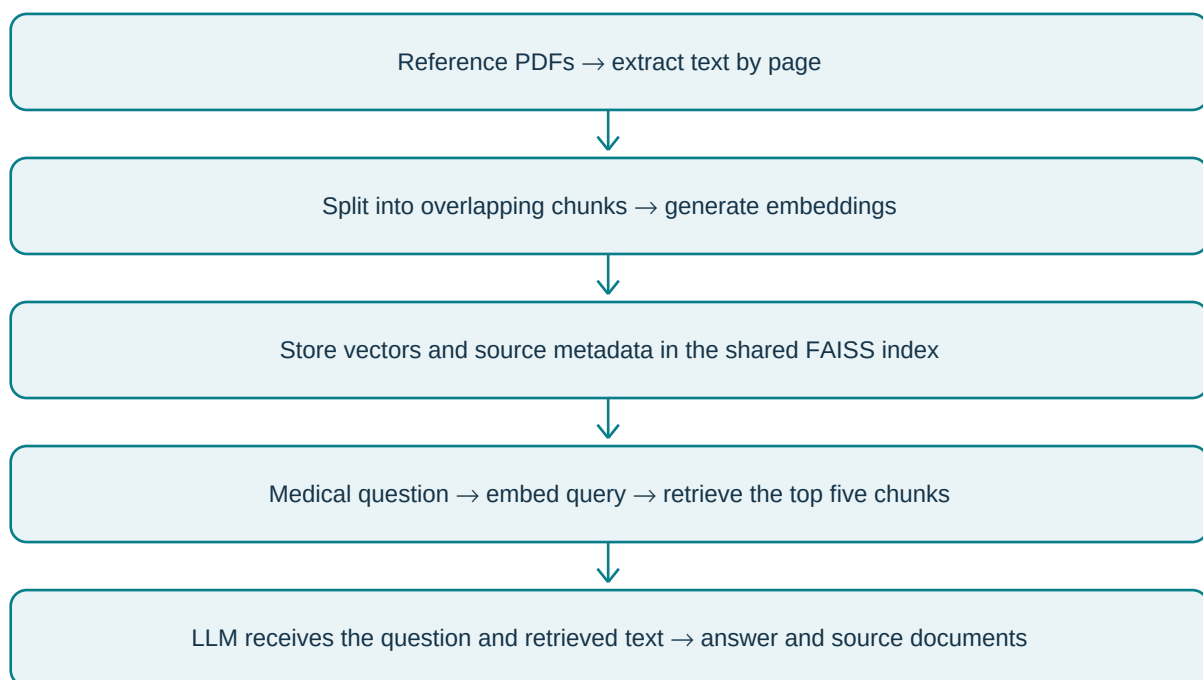
3. How FAISS calculates relevance

For patient summaries, the code normalizes vectors and creates `faiss.IndexFlatIP(vectors.shape[1])`. It performs exact inner-product comparisons against the stored vectors. With normalized vectors, the result is cosine similarity.

similarity = normalized question vector · normalized summary vector

A higher score means greater similarity in embedding space. It is **not a probability that the information is correct**. Exact search is practical for a small collection of patient-summary chunks.

4. Shared reference-document RAG flow



RAG means **Retrieval-Augmented Generation**: retrieve relevant evidence first, then generate an answer using it. The project's RAGChain supplies retrieved chunk text to the LLM and returns source documents.

Reference metadata includes filename, page, chunk index and document ID. These identify retrieved sources; they do not automatically establish that every generated claim is supported.

The cosine-similarity explanation above describes the patient-summary index specifically. Scores from different index configurations should not be treated as interchangeable.

Configuration and BRD example

Implementation defaults and the role of each system component

Detail	Patient summaries	Shared reference PDFs
Purpose	Selected patient context	Uploaded medical references
Default embedding model	text-embedding-3-small	text-embedding-ada-002
Chunk size / overlap	1,800 characters / none	1,000 characters / 200 characters
Index organization	Separate index per patient	Shared reference index
Persistence	patient_summary_vectors table in application SQLite DB	data/embeddings/faiss/ references.sqlite
Extra context	Patient snapshot; record fingerprint	Filename; page; chunk index; document ID

These are code defaults. Environment settings can override configurable values.

5. Applying the BRD sample scenario

"My 70-year-old father has chronic kidney disease. I want to book a nephrologist for him. Also, can you summarize latest treatment methods?"

Task	Responsible component
Identify father and verify access	Patient records and authorization
Retrieve patient context	Patient DB and patient-summary vector memory
Find a slot and book	Doctor Schedule API and scheduling service
Find current treatment information	Trusted external medical search
Retrieve uploaded reference material	Shared reference FAISS index
Compose the response	LLM using context and tool results

The vector database provides relevant context; appointment services perform bookings, and the patient database remains the source of medical records. "Latest" information requires current external sources or refreshed documents.

Sources: reference/healthcare_project_brd.pdf, pp. 3-4; src/config.py; src/vector_store/patient_summaries.py; src/vector_store/faiss_store.py; src/embeddings/embedding_manager.py; src/data_processing/ingestion.py; src/data_processing/text_processing.py; src/chains/rag_chain.py.